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Calibration Audit of Production Astronomical Transient Classifiers on ZTF Alert Streams

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ABSTRACT

Probability estimates from transient classifiers increasingly inform spectroscopic follow-up decisions, but their calibration on real survey data remains insufficiently characterized. We present a calibration audit of three ZTF-era classification systems—ALeRCE's Balanced Random Forest (Förster et al. 2021), Fink's SuperNNova (Möller & de Boissière 2020), and the NEEDLE framework (Sheng et al. 2024)—using spectroscopically confirmed transients from the ZTF Bright Transient Survey (Fremling et al. 2020).

We evaluate calibration via Expected Calibration Error (ECE), reliability diagrams, and Brier scores, and assess temperature scaling (Guo et al. 2017) with cross-validation. ALeRCE is systematically underconfident (ECE = 0.271), reduced to 0.097 after scaling ($T = 0.357$), increasing objects above $p > 0.8$ from 20 to 425 at 91% precision. Fink's SuperNNova produces scores for 64.1% of objects, with non-Ia transients abstaining at 2.6× the SN Ia rate; on the accepted subset, temperature scaling reduces ECE from 0.183 to 0.051 but eliminates all predictions above $p \geq 0.8$, showing that improved ECE need not preserve high-threshold triage utility. NEEDLE shows good aggregate calibration (ECE = 0.048), but classwise analysis reveals severe miscalibration for SLSN-I (73% mean confidence, 31% accuracy), with three silent failures at >99.9% confidence. Temperature scaling worsens calibration in this setting, suggesting class-asymmetric miscalibration that cannot be corrected by a single global rescaling.

These findings demonstrate that calibration audits reveal operationally relevant behavior hidden by accuracy metrics. Direct application to LSST is limited by cadence and depth differences, but the framework is transferable. Code and results are available at https://github.com/pallavi-2000/transient-calibration-audit.

Keywords: methods: statistical — surveys — supernovae: general — techniques: photometric

1. INTRODUCTION

The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) is expected to produce up to several million alerts per night at full survey scale (Ivezić et al. 2019), with official Rubin reports quoting eventual rates of up to approximately 7×10^6 alerts per night, while global spectroscopic follow-up capacity remains fixed at roughly 10^5 spectra per year—a mismatch of approximately two orders of magnitude. The first LSST alerts were issued in 2026 February, with the full survey beginning later this year, making the calibration of broker probability outputs an immediate operational concern. ZTF (Bellm et al. 2019) has served as the primary precursor survey, and the community has developed automated alert brokers to triage this data deluge and identify the most scientifically valuable transients for follow-up.

Several broker systems now operate in production on the ZTF alert stream—ALeRCE (Förster et al. 2021), Fink (Möller et al. 2021), ANTARES (Narayan et al. 2018), and Lasair (Smith et al. 2019)—ingesting alerts in real time and deploying machine learning classifiers to assign class probabilities to each transient. Astronomers routinely use these probability outputs to make follow-up decisions: a transient classified as "90% SN Ia" is far more likely to receive telescope time than one classified at "60% SN Ia." Frameworks such as NEEDLE (Sheng et al. 2024) extend classification to rare transient types including superluminous supernovae (SLSNe) and tidal disruption events (TDEs).

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For broker classifiers that output explicit class probabilities, those scores are routinely reported to end users and increasingly inform spectroscopic follow-up decisions (Förster et al. 2021; Möller et al. 2021; Sheng et al. 2024). This practice implicitly assumes that stated probabilities are calibrated—that is, among all objects assigned a probability p of belonging to a given class, a fraction p actually belongs to that class (Dawid 1982). Whether this assumption holds has not previously been evaluated for production classifiers on real survey data. Miscalibrated probabilities risk misallocated resources: an overconfident classifier may direct follow-up toward objects it misidentifies, while an underconfident one may suppress scientifically valuable candidates below decision thresholds.

The machine learning literature has established that calibration behavior is architecture-dependent: modern neural networks tend toward overconfidence (Guo et al. 2017), while Random Forests exhibit the opposite—systematic underconfidence driven by ensemble averaging (Niculescu-Mizil & Caruana 2005; Boström 2008). Class imbalance corrections further distort probability estimates in ways that vary by class (van den Goorbergh et al. 2022). These properties have direct relevance for broker classifiers but have not been evaluated in this domain.

Two recent works include limited calibration analysis. de Soto et al. (2024) present per-class calibration curves for Superphot+ deployed on the ANTARES broker, noting overconfident SN Ib/c and underconfident SN Ia predictions but without computing formal calibration metrics. Tung (2025) evaluates calibration of ATCAT on simulated ELAsTiCC data, applying post-hoc corrections. However, to our knowledge, no prior work has conducted a systematic calibration audit of production classifiers operating on real survey data using standard metrics across multiple systems.

In this paper, we present, to our knowledge, the first such audit. We evaluate three classifiers across three production systems: ALeRCE's Balanced Random Forest light-curve classifier (`lc_classifier_v1.1.13`; Sánchez-Sáez et al. 2021), Fink's SuperNNova classifier (Möller & de Boissière 2020), and the NEEDLE hybrid CNN+DNN framework (Sheng et al. 2024). We restrict our analysis to classifiers that output explicit class probabilities amenable to calibration analysis; other brokers such as Lasair, ANTARES, and AMPEL employ crossmatching or feature-based approaches outside this scope.[1] Using spectroscopically confirmed transients from the ZTF Bright Transient Survey (BTS; Fremling et al. 2020; Perley et al. 2020) as ground truth, we compute ECE with equal-mass binning (Roelofs et al. 2022), classwise reliability diagrams (Nixon et al. 2019), and Brier scores, then apply temperature scaling with stratified five-fold cross-validation. We find that all three systems exhibit operationally relevant calibration, coverage, or classwise reliability failures, and that temperature scaling is effective for ALeRCE but inappropriate or operationally limited for the other systems.

The paper is organized as follows. Section 2 reviews calibration metrics and post-hoc methods. Section 3 describes data and sample construction. Section 4 presents the audit methodology. Section 5 reports per-classifier findings. Section 6 presents robustness checks. Section 7 discusses architecture-specific calibration patterns and implications for LSST-era broker pipelines. Section 8 states explicit limitations. Section 9 summarizes our conclusions.

2. BACKGROUND

2.1. *Calibration and its Importance*

Calibration describes the alignment between a classifier's stated probabilities and observed empirical frequencies (Dawid 1982). Two related but distinct notions are relevant to this audit. *Top-label confidence calibration* (Guo et al. 2017) requires that predictions made with confidence p are correct with frequency p:

$$\Pr(Y = \hat{Y} \mid \hat{C} = p) = p, \tag{1}$$

where $\hat{Y} = \arg\max_k \hat{P}_k(X)$ and $\hat{C} = \max_k \hat{P}_k(X)$. This is measured by aggregate ECE. *Class-wise calibration* (Nixon et al. 2019) additionally requires, for every class k:

$$\Pr(Y = k \mid \hat{P}_k(X) = p) = p \quad \forall\, k. \tag{2}$$

A classifier can satisfy Equation 1 while violating Equation 2—as we demonstrate for NEEDLE in Section 5. Calibration is distinct from discrimination: high accuracy does not imply good calibration.

2.2. *Calibration Metrics*

Expected Calibration Error (ECE; Naeini et al. 2015) is a standard scalar summary of top-label calibration. For each object i, let $\hat{y}_i = \arg\max_k \hat{p}_{ik}$ denote the predicted class and $\hat{c}_i = \max_k \hat{p}_{ik}$ the associated confidence.

[1] Fink's Random Forest classifier (`rf_snia_vs_nonia`; Leoni et al. 2022), designed for early-time SN Ia detection, returns zero for 93.9% of BTS objects due to operational regime mismatch rather than classifier output; we exclude RF from calibration analysis. See Section 3.

ECE partitions predictions into M bins according to $\hat{c}_i$ and computes the weighted mean absolute difference between empirical accuracy and mean confidence:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \quad (3)$$

where $\mathrm{acc}(B_m) = |B_m|^{-1} \sum_{i \in B_m} \mathbb{1}(\hat{y}_i = y_i)$ and $\mathrm{conf}(B_m) = |B_m|^{-1} \sum_{i \in B_m} \hat{c}_i$. For binary classifiers, $\hat{c}_i = \max(p_i, 1 - p_i)$ and $\hat{y}_i = \mathbb{1}(p_i \geq 0.5)$. Equal-width binning can leave bins near-empty for classifiers with skewed confidence distributions; Roelofs et al. (2022) showed that equal-mass binning yields lower-bias estimates. We adopt equal-mass binning with $M = 15$ throughout; bin-count sensitivity is reported in Appendix C.

Because standard ECE considers only the top-predicted class, it can mask miscalibration in individual class probabilities (Nixon et al. 2019). For multiclass classifiers we therefore report per-class reliability diagrams and classwise ECE. For the binary Fink SuperNNova classifier we report conditional calibration on the accepted set and analyse class-dependent abstention separately. We additionally report the multiclass Brier score (Brier 1950) for ALeRCE and NEEDLE and the standard binary Brier score for Fink SuperNNova as proper scoring rules capturing both calibration and probabilistic sharpness. Note that the Brier score reference value for random guessing is $(K-1)/K$: $\sim$0.5 for binary classifiers and $\sim$0.75 for four-class.

2.3. *Temperature Scaling*

Temperature scaling (Guo et al. 2017) is a single-parameter post-hoc calibration method that rescales the logit vector $\mathbf{z}$ by a learned temperature $T > 0$:

$$\hat{q}_k = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)}. \quad (4)$$

When $T > 1$ the distribution is softened, reducing overconfidence; when $T < 1$ it is sharpened, reducing underconfidence. The temperature is fitted by minimizing negative log-likelihood on a held-out calibration set, as NLL is a strictly proper scoring rule (Gneiting & Raftery 2007). For classifiers that do not produce native logits, vote-fraction probabilities are converted to pseudo-logits via $z_k = \log \hat{p}_k$ prior to scaling; the implications of this approximation are discussed in Section 4.

Temperature scaling applies a single global transformation and therefore cannot correct class-asymmetric miscalibration. Kull et al. (2019) showed that it can produce well-calibrated aggregate reliability diagrams while individual classes remain miscalibrated—precisely the failure mode we document for NEEDLE in Section 5.

2.4. *Calibration of Ensemble Classifiers and Class Weighting*

Random Forests produce systematically underconfident predictions due to ensemble averaging, which biases probabilities toward 0.5 (Niculescu-Mizil & Caruana 2005; Boström 2008). Balanced Random Forests further distort probability estimates by altering the effective class prior through majority-class undersampling, producing systematic overestimation of minority-class probabilities (van den Goorbergh et al. 2022)—the calibration profile we observe for ALeRCE in Section 5. Inverse-frequency class weighting, as employed by NEEDLE, produces the same minority-class inflation and additionally induces class-asymmetric miscalibration that global temperature scaling cannot resolve (van den Goorbergh et al. 2022; Kull et al. 2019).

3. DATA

3.1. *Spectroscopic Ground Truth*

We use the ZTF Bright Transient Survey (BTS; Fremling et al. 2020; Perley et al. 2020) as our source of spectroscopic ground truth. BTS provides systematic spectroscopic classifications for bright ZTF transients, yielding 7,167 objects with confirmed type assignments. Because BTS is magnitude-limited and spectroscopically selected, the evaluation set represents a bright, confirmed-transient sample rather than a population-representative sample of all ZTF alerts.

3.2. *Label Harmonization*

BTS spectroscopic subtypes are mapped into the class labels used across all three classifiers: SN Ia and its subtypes (Ia-91T, Ia-91bg, Ia-02cx, Ia-CSM) map to SN Ia; SN Ib, Ic, Ic-BL, Ib/c, and Ibn map to SN Ibc; SN II, IIn, IIb, and IIP map to SN II; SLSN-I and SLSN-II map to SLSN; and TDE maps to TDE. Objects with types absent from all classifier taxonomies (novae, LRN, LBV) are excluded, yielding 7,116 mapped objects. For classifiers without a TDE output class, TDE objects are excluded from that classifier-specific analysis.

3.3. *Stratified Evaluation Sample*

Rare classes constitute fewer than 3% of the 7,116 mapped BTS objects. To ensure adequate representation for per-class calibration diagnostics, we construct a stratified sample capping SN Ia at 600 objects and retaining all available objects for rarer classes, yielding 1,436 objects (Table 1).

This sample is designed to support per-class reliability analysis, not to estimate population-averaged ECE. Aggregate ECE values should therefore be interpreted

Table 1. Stratified Evaluation Sample Composition

Class	Count	Fraction
SN Ia	600	41.8%
SN II	341	23.7%
SN Ibc	225	15.7%
SLSN	97	6.8%
TDE	39	2.7%
Other	134	9.3%
Total	1436	100%

for this evaluation distribution; per-class reliability diagrams are the primary diagnostic for class-specific calibration behavior (Guo et al. 2017).

3.4. *Classifier Prediction Retrieval*

ALeRCE. We query the ALeRCE API[2] for probabilities from `lc_classifier_v1.1.13` (Sánchez-Sáez et al. 2021), a Balanced Random Forest returning a 15-class probability vector spanning transient and variable-star categories. Predictions are retrieved for 1,149 of 1,436 objects. ALeRCE's production taxonomy does not include a TDE class, so the 35 retrieved TDE objects are excluded, leaving 1,114 objects for analysis across four transient classes: SN Ia, SN Ibc, SN II, and SLSN. Probability vectors are renormalized to sum to unity over this four-class subspace; robustness of this restriction is assessed in Section 6.

Fink. We query the Fink API[3] for SuperNNova (`snn_snia_vs_nonia`; Möller & de Boissière 2020), a recurrent neural network returning a scalar score $\in [0, 1]$ representing $P(\text{SN Ia})$ for objects satisfying its photometric eligibility criteria. We retrieve the most recent alert per object, obtaining predictions for 1,237 objects. A score of exactly zero indicates eligibility failure rather than $P(\text{SN Ia}) = 0$; the treatment of zero-valued scores is described in Section 4.4.[4]

NEEDLE. We cloned the publicly available repository[5] and downloaded the public dataset (`data.hdf5`) from the associated Kaggle release (Sheng et al. 2024). Local inference was performed using the five pre-trained models from the `lasair_th_r` family via `src/needle_extraction.py`. Each model was trained with an independently drawn held-out test set; we resolve ZTF object positions via `hash_table.json` rather than the stale indices in `testset_obj.json`. This yields 429 predictions across 278 unique objects in three classes (SN, SLSN-I, TDE). Because each model has an independent held-out set, 43 objects appear in two or more models' test sets and receive multiple predictions. Our primary analysis deduplicates to the object level by averaging probabilities across models (278 unique objects); the 429-prediction model-instance set is retained as a sensitivity analysis (Section 5.3.4). The 100% inter-model class agreement across all 43 multi-model objects validates the averaging methodology.

3.5. *Retrieval Completeness*

Broker predictions were retrieved for 1,149 of 1,436 BTS objects (ALeRCE; 80.0%) and 1,237 (Fink SuperNNova; 86.1%). To verify that missing predictions do not introduce systematic bias, we tested whether retrieval is class-dependent and whether retrieved and missing objects differ on observable BTS properties.

ALeRCE retrieval is significantly class-dependent (χ^2 $p = 4.1 \times 10^{-4}$): SN Ibc transients are most frequently missing (25.0%), consistent with the light-curve classifier's minimum-detection requirements (Sánchez-Sáez et al. 2021). Retrieved objects have significantly longer median durations than missing ones (26.3 versus 24.4 days). ALeRCE calibration metrics should therefore be interpreted as conditional on the retrieval-eligible subset of BTS-bright transients.

Fink's API returns a score for every submitted alert, so retrieval shows no significant class-dependence (χ^2 $p = 0.061$). Note that Fink *coverage*—the fraction of retrieved objects with non-zero scores—is strongly class-dependent and is analyzed separately in Section 5.2.1; retrieval and coverage are distinct quantities.

4. METHODS

4.1. *Evaluation Regimes*

The three systems differ in output taxonomy, prediction coverage, and probability space, so calibration is assessed under classifier-specific estimands. For ALeRCE, we evaluate four-class transient-subspace calibration (SN Ia, SN Ibc, SN II, SLSN) after renormalizing the native 15-class probability vector. For Fink SuperNNova, we evaluate conditional binary calibration on the accepted set—the subset of objects for which the classifier emits a non-zero score. For NEEDLE, the primary analysis is object-level three-class calibration (SN,

[2] https://api.alerce.online

[3] https://api.fink-portal.org

[4] Fink's Random Forest classifier (`rf_snia_vs_nonia`; Leoni et al. 2022), designed for early-time SN Ia detection, returns zero for 93.9% of BTS objects reflecting operational regime mismatch rather than classifier output; we exclude RF from calibration analysis.

[5] https://github.com/XinyueSheng2019/NEEDLE

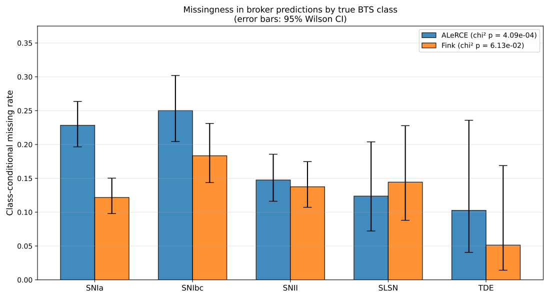

Figure 1. Class-conditional missing rates in broker predictions on the 1,436-object BTS evaluation sample. Error bars are 95% Wilson confidence intervals. ALeRCE retrieval is significantly class-dependent (χ^2 $p = 4 \times 10^{-4}$), with SN Ibc most likely to be missed; Fink retrieval is approximately uniform across classes (χ^2 $p = 0.06$).

SLSN-I, TDE), with model-instance counting retained as a sensitivity analysis.

These regimes define the estimand for each reported ECE value. Classifier-specific ECE values quantify calibration within the corresponding evaluation population and label space and are not directly interchangeable as population-level calibration errors. We compare failure modes across systems rather than ECE magnitudes.

4.2. *Calibration Metrics*

For each object i, let $\hat{\mathbf{p}}_i$ denote the classifier probability vector over the evaluated label space, y_i the ground-truth label, $\hat{y}_i = \arg\max_k \hat{p}_{ik}$ the predicted class, and $\hat{c}_i = \max_k \hat{p}_{ik}$ the associated confidence. For binary classifiers, $\hat{c}_i = \max(p_i, 1 - p_i)$ and $\hat{y}_i = \mathbb{1}(p_i \geq 0.5)$.

We compute top-label ECE using equal-mass binning with $M = 15$ bins:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \qquad (5)$$

where $\mathrm{acc}(B_m) = |B_m|^{-1} \sum_{i \in B_m} \mathbb{1}(\hat{y}_i = y_i)$ and $\mathrm{conf}(B_m) = |B_m|^{-1} \sum_{i \in B_m} \hat{c}_i$. Equal-mass binning is adopted because several classifiers have highly concentrated confidence distributions; equal-width binning and alternative bin counts are evaluated as sensitivity checks (Appendix C).

We additionally report the multiclass Brier score (Brier 1950) for ALeRCE and NEEDLE and the standard binary Brier score for Fink SuperNNova as proper scoring rules capturing both calibration and probabilistic sharpness.

4.3. *Classwise Calibration*

Aggregate top-label ECE can conceal class-specific probability errors. We therefore compute classwise calibration by treating each class k as a one-versus-rest probabilistic prediction: predictions are binned by $\hat{p}_{ik}$ and compared against $\mathbb{1}(y_i = k)$, yielding a classwise reliability curve and classwise ECE for each evaluated class (Nixon et al. 2019). Classwise analysis is the primary diagnostic for NEEDLE, where aggregate ECE conceals opposing miscalibration directions across classes.

4.4. *Treatment of Abstentions*

Fink SuperNNova operates as a selective classifier (Geifman & El-Yaniv 2017): it emits a probability estimate only for objects satisfying its photometric eligibility criteria, returning exactly zero otherwise. We therefore evaluate SuperNNova within a selective classification framework, distinguishing two quantities: *coverage*—the fraction of retrieved objects with non-zero scores (64.1% of the 1,237 retrieved objects)—and *conditional calibration*—ECE computed on the accepted subset. Retrieval completeness is analyzed separately in Section 3.5.

We additionally quantify abstention bias—the difference between accepted-set calibration and calibration reweighted to the BTS class prior—to estimate how selective abstention affects the population-equivalent calibration gap. A positive abstention bias indicates the conditional measurement underestimates population-level miscalibration.

4.5. *Temperature Scaling*

We fit a scalar temperature $T \in [0.01, 10.0]$ by minimizing negative log-likelihood on held-out data using bounded scalar optimization. For classifiers producing native logits, temperature scaling is applied directly to the logit vector. For the ALeRCE Balanced Random Forest, which produces vote-fraction probabilities rather than logits, we apply the pseudo-logit transformation $z_k = \log \hat{p}_k$ prior to scaling. This is a practical heuristic rather than a theoretically grounded transformation; results are interpreted empirically and compared against unscaled probabilities in Section 6.

Temperature scaling is evaluated with stratified five-fold cross-validation: class ratios are preserved across folds, T is fitted on four folds, and ECE is evaluated on the held-out fold, with results averaged across folds.

We interpret temperature scaling both as a post-hoc correction and as a structural diagnostic. A finite T that improves ECE suggests approximately uniform miscalibration correctable by rescaling. A temperature at the optimization boundary, a worsened held-out ECE, or inconsistent classwise changes indicates structural miscalibration that scalar rescaling cannot resolve.

For classifiers where global temperature scaling is inappropriate, we additionally evaluate per-class temperature scaling (Fang & Orabona 2020), which fits an independent T_k for each class.

4.6. *Operational Threshold Analysis*

Broker probabilities are commonly applied through threshold-based selection rules. We therefore evaluate the operational consequences of calibration at fixed probability thresholds τ, reporting the number of objects satisfying $\hat{c}_i \geq \tau$ and the precision among selected objects, before and after temperature scaling. For binary classifiers, thresholds are applied to P(SN Ia); for multiclass classifiers, to the top-label confidence $\hat{c}_i$. This distinguishes formal ECE improvements from changes that are operationally useful for spectroscopic triage.

4.7. *Uncertainty Estimation*

Uncertainty intervals for ECE and calibration gaps are estimated with 1,000 bootstrap resamples of the evaluation objects. For object-level analyses, resampling is performed over unique objects; for model-instance sensitivity analyses, over prediction instances. For binomial quantities—classwise accuracy, coverage, and abstention rates—we report 95% Wilson confidence intervals.

5. RESULTS

Table 2 summarizes the calibration outcomes across all three systems. The classifiers exhibit distinct reliability failure modes. ALeRCE is systematically underconfident, with probability mass compressed below high-confidence decision thresholds. Fink SuperNNova behaves as a selective classifier: it emits calibrated probabilities only on an accepted subset and abstains non-uniformly across classes. NEEDLE appears well calibrated in aggregate, but classwise analysis reveals rare-class overconfidence hidden by the top-label ECE. We report each result below.

5.1. *ALeRCE: Systematic Underconfidence*

The ALeRCE light curve classifier exhibits systematic underconfidence across all transient classes (Figure 2). The overall accuracy is 0.742, but the mean confidence is only 0.471—a gap of +0.271 indicating that the classifier is substantially more accurate than its confidence scores suggest. The reliability diagram shows all bins lying above the diagonal, with confidence values compressed into the range $[0.3, 0.75]$ and never exceeding 0.75.

This underconfidence pattern is consistent with known ensemble calibration behavior; we discuss the mechanism in Section 7.1.

Per-class analysis (Figure 3) reveals that underconfidence is pervasive: SN Ia (ECE = 0.195, gap = +0.335), SN Ibc (ECE = 0.154, gap = +0.201), and SN II (ECE = 0.150, gap = +0.393) all exhibit strong underconfidence. SLSN is a notable exception with a slight overconfidence (gap = −0.055), potentially reflecting the Balanced Random Forest's inflation of minority-class probabilities (van den Goorbergh et al. 2022).

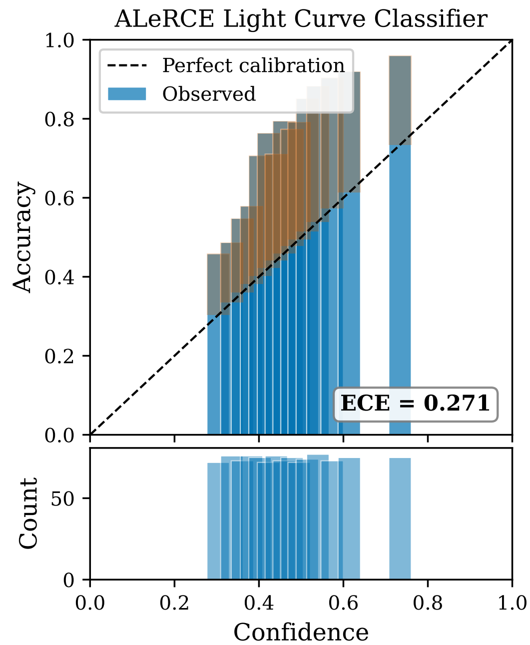

Figure 2. Reliability diagram for the ALeRCE light curve classifier showing systematic underconfidence: all bins lie above the perfect calibration diagonal. Confidence values are compressed into the range $[0.3, 0.75]$. ECE = 0.271 with equal-mass binning (15 bins).

Temperature scaling with $T = 0.357 \pm 0.012$ (stratified five-fold CV) reduces the cross-validated ECE from 0.276 to 0.097—a 65% improvement (Figure 4), increasing objects exceeding $p > 0.8$ from 20 to 418 at 90.7% precision in pooled held-out cross-validation (full-sample: 425 objects at 91.1% precision; Figure 14). The sharpening effect ($T < 1$) is the correct direction for underconfident classifiers, spreading the compressed confidence range $[0.3, 0.75]$ out to $[0.35, 1.0]$. We note that $T = 0.357$ is more aggressive than values typically reported in the neural network calibration literature (where $T > 1$ is standard for overconfident models), reflecting the severity of RF underconfidence when distributed across 15 classes. The non-cross-validated ECE of 0.030 shown in Figure 4 illustrates the visual effect of calibration but overestimates improvement due to fitting and evaluating on the same data; the cross-validated estimate of 0.097 is the appropriate value for reporting.

Table 2. Calibration Audit Summary

Classifier	Architecture	Evaluation Regime	N	ECE	Brier	Post-hoc Method	$\mathrm{ECE}_{\mathrm{post}}$
ALeRCE	BRF	4-class transient subspace	1114	0.271 $[0.249, 0.296]$	0.486	Temp. ($T = 0.357$)	0.097
Fink RF†	RF	—	1237	—	—	N/A (regime mismatch)	—
Fink SNN‡ (cond.)	RNN	Binary SN Ia, accepted set	793	0.183 $[0.154, 0.220]$	0.274	Temp. ($T = 3.65$)	0.051
NEEDLE (obj.-level)§	CNN+DNN	3-class object-level	278	0.048 $[0.048, 0.111]$	0.265	Global T worsens	0.169
(model-instance)			429	0.073 $[0.063, 0.117]$	0.241	(sensitivity; §5.3.4)	—

Note—ECE computed with equal-mass binning (15 bins). 95% confidence intervals via bootstrap (1,000 replicates). Post-hoc ECE for ALeRCE is the cross-validated estimate (stratified 5-fold). Brier score: 0 is perfect; for reference, random guessing gives ∼0.5 for binary classifiers and ∼0.75 for 4-class. †Fink RF abstains on 93.9% of BTS objects with near-uniform abstention across classes (regime mismatch; see §3); excluded from calibration analysis. ‡Fink SNN conditional metrics on 793 non-zero predictions (64.1% coverage). Reweighting to BTS prevalence yields population-equivalent gap of 0.460 (abstention bias +0.024). Post-T value of 0.051 is operationally null at $p \geq 0.8$; see Table 3 and §5.2.1. §NEEDLE object-level (primary): 278 unique objects, deduplicated by averaging probabilities across models. See §5.3 and §6.

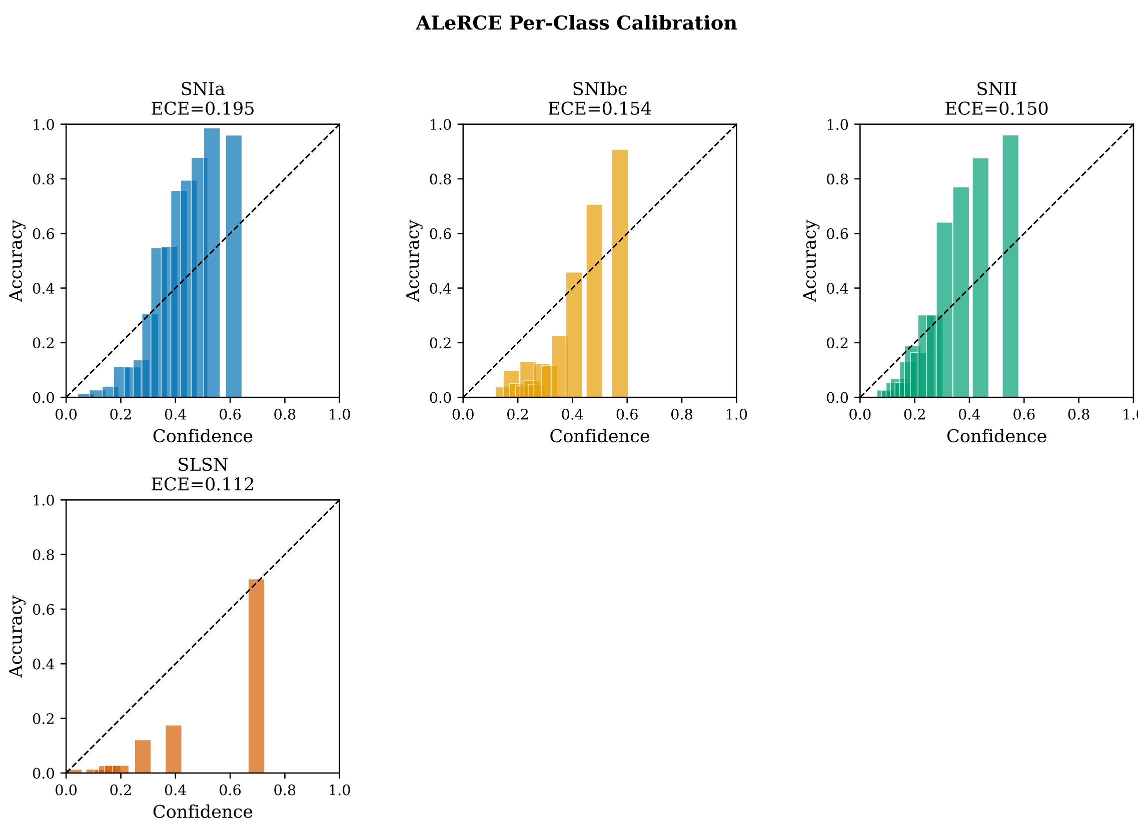

Figure 3. Per-class reliability diagrams for the ALeRCE classifier. SN Ia, SN Ibc, and SN II show consistent underconfidence (bars above diagonal). SLSN shows slight overconfidence, potentially due to minority-class probability inflation from the Balanced Random Forest architecture.

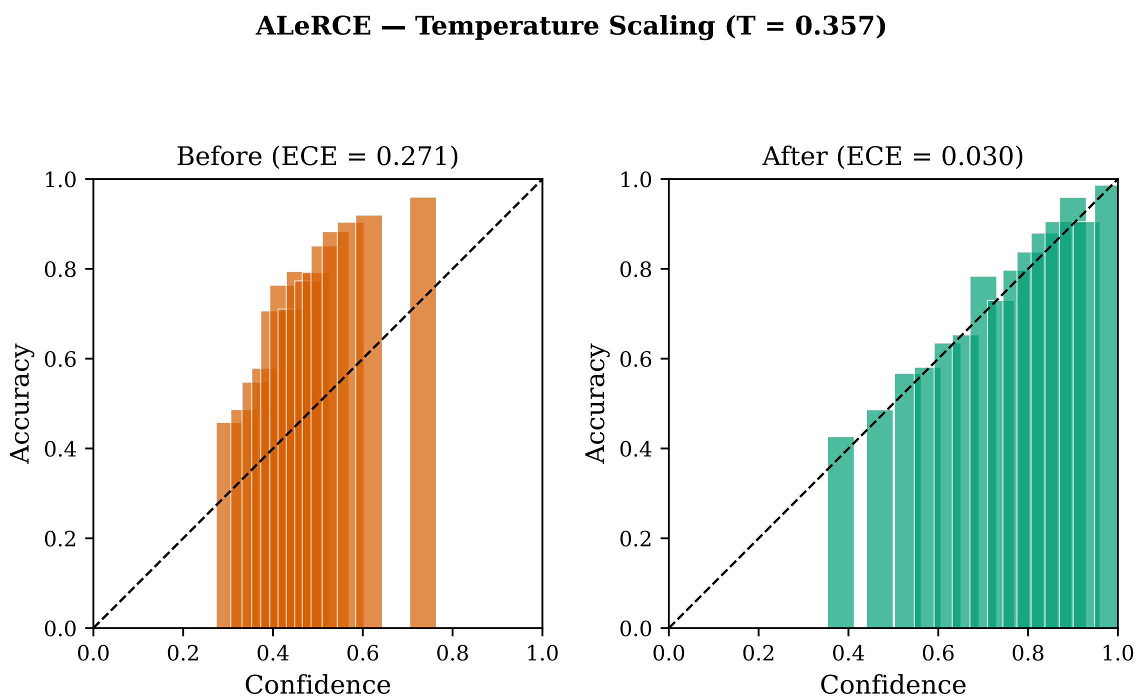

Figure 4. ALeRCE reliability diagrams before (left) and after (right) temperature scaling with $T = 0.357$. The displayed ECE of 0.030 is computed on the full dataset; the cross-validated estimate is 0.097 (65% improvement). The sharpening effect spreads the compressed confidence distribution to span the full $[0.35, 1.0]$ range.

5.2. *Fink: Selective Classification*

Fink SuperNNova operates as a selective classifier (Geifman & El-Yaniv 2017), as established in Section 4.4: it emits a probability estimate only for objects satisfying photometric eligibility criteria. We evaluate SuperNNova within this selective classification framework.

5.2.1. *SuperNNova: Selective Classification with Class-Dependent Abstention*

Fink's SuperNNova classifier (Möller & de Boissière 2020) emits non-zero scores for 793 of 1,237 BTS objects (64.1% coverage). Abstention is strongly class-dependent (χ^2, $p = 8.4 \times 10^{-37}$): SN Ia objects abstain at 19.0%, SN Ibc at 29.8%, SN II at 55.7%, TDE at 62.2%, and SLSN at 67.5% (Figure 5). The monotonic increase from bright thermonuclear to faint core-collapse and rare classes is consistent with SuperNNova's photometric data-quality requirements: rarer, fainter classes have sparser light curves that more frequently fail the minimum-photometry gate. Non-SN Ia transients abstain at 2.6 times the rate of SN Ia (48.5% versus 19.0%), and the hard gap from exactly zero to the first non-zero score (0.0051) is diagnostic of an explicit gate rather than a continuous low-confidence regime (Figure 8).

Conditional calibration. On the 793 accepted objects, the classifier achieves overall accuracy 0.589 at threshold $p \geq 0.5$ with a conditional ECE of 0.183 [95% CI: 0.154, 0.220] and Brier score 0.274. Per-class analysis on the accepted set reveals that SuperNNova systematically issues moderate probabilities of SN Ia for non-Ia transients: mean predicted $P(\text{SN Ia})$ is 0.51 for SN II, 0.45 for SN Ibc, 0.68 for SLSN, and 0.73 for TDE objects that pass its eligibility gate, despite the true SN Ia fraction in each class being zero (Figure 6). This is the dominant source of conditional miscalibration.

Abstention bias. The conditional ECE measures calibration on a sample whose class composition differs from BTS: SN Ia is 53.8% of the accepted set versus 42.6% of BTS, while SN II drops from 27.9% to 19.3% and SLSN drops from 6.7% to 3.4% (Figure 7). Reweighting per-class calibration gaps to BTS prevalence yields a population-equivalent weighted-gap of 0.460 versus the conditional weighted-gap of 0.436, an absolute increase of 0.024. This +2.4 percentage-point bias quantifies how selective abstention masks population-level miscalibration.

Temperature scaling. On the accepted set, temperature scaling fits $T = 3.65$ (no bound hit) and reduces conditional ECE to 0.051—a 72% reduction. However, the operational consequence of $T = 3.65$ is fundamentally different from the ALeRCE case (Section 5.1, $T =$ 0.357 sharpening). Because $T > 1$ compresses scores toward 0.5, the post-scaling distribution is restricted to [0.191, 0.708] (versus [0.005, 0.962] pre-scaling). Table 3 shows the consequences for high-threshold selection: pre-calibration, 209 objects pass $p \geq 0.8$ at 73.2% precision and 66 pass $p \geq 0.9$ at 74.2% precision; post-calibration, zero objects pass either threshold because the maximum post-scaled score is 0.708. The 72% ECE reduction reflects the classifier honestly expressing its inherent uncertainty rather than acquiring new discriminative power; classification decisions at $p \geq 0.5$ are unchanged (495 predicted SN Ia in both cases, identical 60.2% precision).

Table 3. Operational consequences of temperature scaling for Fink SuperNNova on the accepted set ($N = 793$). Temperature $T = 3.65$ compresses the score distribution from [0.005, 0.962] to [0.191, 0.708]. At $p \geq 0.5$, decisions are unchanged. At higher thresholds, post-calibration eliminates all predictions while pre-calibration retains operationally useful candidates.

Threshold	Pre-T (raw)		Post-T ($T = 3.65$)	
$p \geq$	N pass	Precision	N pass	Precision
0.5	495	0.602	495	0.602
0.6	448	0.607	188	0.745
0.7	355	0.639	2	1.000
0.8	209	0.732	0	—
0.9	66	0.742	0	—

For *probability reporting* to astronomers, $T = 3.65$ produces honest output. For *high-precision triage*, pre-calibration scores remain operationally useful (209 candidates at 73% precision) despite their formal miscalibration. The contrast between underconfident classifiers (ALeRCE, where calibration unlocks high-confidence selection) and overconfident classifiers (Fink SNN, where calibration eliminates high-confidence selection) is a substantive finding of this audit (Section 7.5).

5.3. *NEEDLE: Object-Level Calibration and Class Imbalance*

5.3.1. *Aggregate Calibration: Object-Level Results*

Our primary NEEDLE analysis evaluates 278 unique ZTF objects, deduplicated by averaging probabilities from all models that classified each object (Section 6). At the object level, NEEDLE achieves aggregate ECE = 0.048 [95% CI: 0.048, 0.111] and overall accuracy 80.2% (Brier = 0.265). This aggregate ECE lies at

Figure 5. Fink SuperNNova class-conditional abstention rates on the BTS evaluation sample ($N = 1237$). Error bars show 95% Wilson confidence intervals. Abstention is strongly class-dependent (χ^2, $p = 8.4 \times 10^{-37}$), with monotonic increase from bright thermonuclear (SN Ia, 19.0%) to faint core-collapse and rare classes (SLSN, 67.5%; TDE, 62.2%). The dashed line shows the overall abstention rate of 35.9%.

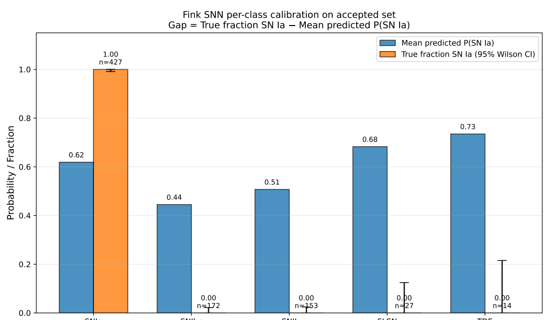

Figure 6. Fink SuperNNova per-class calibration on the accepted set (non-zero predictions, $n = 793$). Blue bars show mean predicted P(SN Ia) and orange bars show the true SN Ia fraction (with 95% Wilson CI). The classifier issues moderate P(SN Ia) values (0.45–0.73) for non-Ia transients that pass its eligibility gate, despite the true SN Ia fraction being zero in those classes. This is the dominant source of conditional miscalibration.

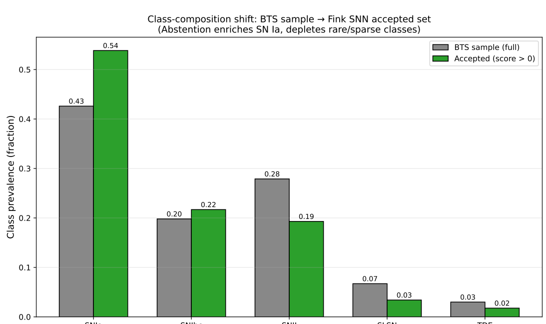

Figure 7. Class-composition shift induced by Fink SuperNNova's selective abstention. The accepted set is enriched in SN Ia (+11.2%) and depleted in SN II (−8.6%) and SLSN (−3.3%) relative to the BTS sample. This biases the conditional ECE estimate by approximately +0.024 when reweighted to BTS prevalence (Section 5.2.1).

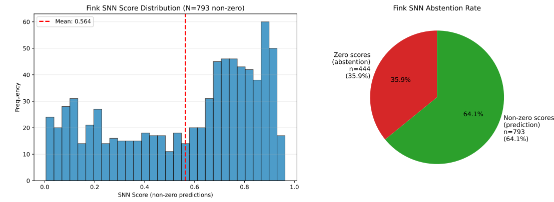

Figure 8. Zero-score distribution for Fink SuperNNova. Left: full score distribution with 444 objects (35.9%) at exactly zero and a hard gap of 0.005 to the first non-zero score. Right: non-zero score distribution for SN Ia and non-SN Ia objects, showing active discrimination when predictions are made (mean = 0.564, median = 0.664).

Table 4. NEEDLE Per-Class Calibration (Object-Level, $N = 278$)

Class	n_{true}	n_{pred}	Accuracy	Mean Conf.	Gap	Direction
SN	242	193	0.984	0.865	+0.119	Underconfident
SLSN-I	15	39	0.308	0.730	−0.423	Overconfident
TDE	21	46	0.457	0.752	−0.296	Overconfident

NOTE—Gap = Accuracy − Mean Confidence (when class is predicted). Positive gap: classifier is more accurate than confident (underconfident). Negative gap: classifier is more confident than accurate (overconfident). n_{pred} exceeds n_{true} for SLSN-I and TDE because many non-rare objects are misclassified into those categories.

the boundary of the heuristic "well-calibrated" criterion (ECE < 0.05), a markedly better result than the model-instance ECE of 0.073 that inflates rare-class contributions due to repeated appearances across held-out sets (Section 5.3.4).

Figure 11 shows the object-level reliability diagram. The aggregate bars track the diagonal reasonably well, consistent with the low ECE. However, as the per-class analysis below reveals, this aggregate view conceals severe, class-specific miscalibration driven by class imbalance.

5.3.2. *Per-Class Miscalibration and Class Imbalance*

Table 4 and Figure 12 show the classwise calibration decomposition at the object level. The three classes exhibit fundamentally different calibration behavior:

SN (majority class, $n_{\text{true}} = 242$). SN predictions are underconfident: when NEEDLE predicts SN, it achieves accuracy 0.984 [Wilson 95% CI: 0.955, 0.995] but assigns mean confidence 0.865, yielding a gap of +0.119 [bootstrap 95% CI: +0.091, +0.150]. This is consistent with the systematic suppression of majority-class probabilities under inverse-frequency weighting, which upweights rare-class gradients at the expense of the dominant class. The CI does not include zero, and the magnitude is well-resolved relative to the CI width.

SLSN-I (rare class, $n_{\mathrm{true}} = 15$). SLSN-I predictions are overconfident in direction with magnitudes that carry substantial uncertainty given the small sample size. When NEEDLE predicts SLSN-I, it achieves accuracy 0.308 [Wilson 95% CI: 0.186, 0.464] but assigns mean confidence 0.730, yielding a gap of -0.423 [bootstrap 95% CI: -0.520, -0.312]. The CI does not include zero, so the direction (overconfident) is robust; however, model-instance sensitivity analysis (Section 5.3.4) yields a smaller gap of -0.156 [-0.231, -0.088] on $n_{\mathrm{pred}} = 87$ predictions, indicating that the magnitude is sample-size sensitive. Of the 39 objects predicted as SLSN-I at the object level, 24 are false positives.

TDE (rare class, $n_{\mathrm{true}} = 21$). TDE predictions are also overconfident in direction at the object level: accuracy 0.457 [Wilson 95% CI: 0.322, 0.598], mean confidence 0.752, gap -0.296 [bootstrap 95% CI: -0.413, -0.172]. The model-instance sensitivity analysis yields a markedly smaller gap of -0.021 [-0.078, $+0.035$] on $n_{\mathrm{pred}} = 131$ predictions, with a CI that crosses zero. We therefore report TDE overconfidence as *directionally consistent* between object-level and model-instance analyses, but conclude that the magnitude is *not robustly resolved* at the present sample size.

The asymmetry—SN underconfident while both rare classes are overconfident—is the canonical signature of inverse-frequency class weighting identified by van den Goorbergh et al. (2022). The aggregate ECE of 0.048 conceals this because the majority-class underconfidence (SN, $n = 242$) partially cancels the minority-class overconfidence (SLSN-I + TDE, $n = 36$) in the weighted average. We emphasise that the rare-class point estimates are sample-size constrained: bootstrap intervals on the SLSN-I gap are ± 0.10 in width and on the TDE gap ± 0.12 in width (Section 5.3.4). The qualitative direction of the asymmetry—SN underconfident, rare classes overconfident—is robust; the precise magnitudes should be interpreted as strongly suggestive of class-imbalance-induced miscalibration rather than as definitive quantitative measurements.

5.3.3. *Silent Failures and Confidence-Accuracy Mismatch*

At the object level, five objects are classified with $\geq$90% confidence but are incorrect (silent failures, 1.8% of 278 objects). Table 5 lists these objects.

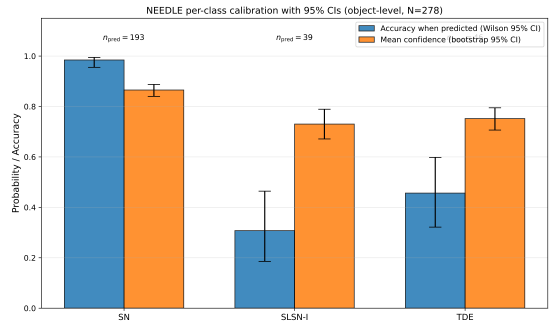

Figure 9. NEEDLE per-class calibration with explicit 95% confidence intervals (object-level analysis, $N = 278$). Blue bars show accuracy when each class is predicted, with Wilson 95% binomial CIs; orange bars show mean confidence assigned to the predicted class with bootstrap 95% CIs. Direction of miscalibration is robust for all three classes (SN underconfident, SLSN-I and TDE overconfident); magnitudes for rare classes ($n_{\mathrm{pred}} = 39$ for SLSN-I, $n_{\mathrm{pred}} = 46$ for TDE) carry substantial uncertainty.

Table 5. NEEDLE Object-Level Silent Failures ($\geq$90% confidence, wrong)

ZTF ID	Predicted	True	Confidence
ZTF20aagikvv	SN	SLSN-I	1.000
ZTF21aakjkec	SN	SLSN-I	0.9997
ZTF19abzoyeg	SN	SLSN-I	0.999
ZTF21aagnwkk	TDE	SN	0.901
ZTF21aautijg	SLSN-I	SN	0.907

Note—Three of five silent failures are true SLSN-I objects misclassified as SN at near-certainty confidence. This pattern is consistent with the class-imbalance mechanism: rare objects are predicted as majority class with extreme confidence when they fall outside the learned minority-class decision region.

Three of the five silent failures (ZTF20aagikvv, ZTF21aakjkec, ZTF19abzoyeg) are true SLSN-I objects classified as SN with confidence > 0.998. These objects fall outside NEEDLE's learned SLSN-I decision boundary despite their spectroscopic confirmation; the inverse-frequency weighting inflates the SLSN-I probabilities for objects *within* the boundary while leaving ambiguous cases assigned entirely to the SN class with extreme confidence. For spectroscopic triage, a near-certainty SN prediction from NEEDLE carries an embedded risk: three independently spectroscopically confirmed SLSN-I objects were classified this way.

5.3.4. *Model-Instance Sensitivity: Robustness Across Held-Out Models*

As a sensitivity analysis, we evaluate all 429 model-instance predictions, treating each model's prediction on its held-out test objects as independent (see Section 6 for the deduplication methodology). The model-instance

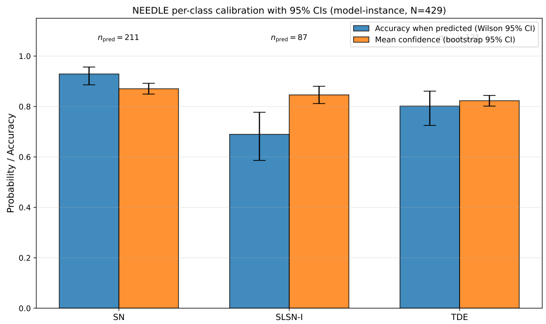

Figure 10. NEEDLE per-class calibration with 95% confidence intervals under the model-instance counting convention ($N = 429$). Compare with Figure 9 (object-level, $N = 278$). Direction of miscalibration is consistent with the object-level analysis for SN and SLSN-I; the TDE gap shrinks substantially in the model-instance counting and the bootstrap CI crosses zero, illustrating that magnitude estimates for rare classes are sample-size sensitive and should be interpreted within the bracket spanned by both counting conventions.

aggregate ECE is 0.073 [0.063, 0.117], with 17 silent failures (4.0%).

The higher model-instance ECE (0.073 vs. 0.048 object-level) reflects an artifact of the evaluation design: rare-class objects (SLSN-I, TDE) appear in many more models' held-out test sets than SN objects (mean 5.0 models per rare object vs. 1.03 for SN), inflating the effective rare-class sample weight and shifting the aggregate ECE upward. The 100% inter-model class agreement across all 43 multi-model objects—zero objects show disagreement in predicted class across models—validates that the averaging methodology used in the primary analysis is sound.

The model-instance analysis yields per-class gaps with 95% bootstrap intervals of +0.059 [+0.012, +0.100] for SN, −0.156 [−0.231, −0.088] for SLSN-I, and −0.021 [−0.078, +0.035] for TDE. The direction of miscalibration is consistent with the object-level analysis (Section 5.3.2) for SN and SLSN-I, and the TDE direction is consistent at the object level but with a model-instance CI that crosses zero. The smaller magnitude differences between object-level and model-instance analyses for rare classes reflect the rare-class prediction count: rare-class objects appear repeatedly across multiple held-out test sets in the model-instance analysis (mean 5.0 instances per rare-class object versus 1.03 for SN), inflating the effective n_{pred}. Both analyses agree on direction; the precise magnitude depends on whether predictions are deduplicated to the object level or not, and on which counting convention is more appropriate for the science use-case.

5.3.5. *Interpretation: Class Imbalance and Learned Biases*

The class-asymmetric miscalibration we observe is consistent with the expected effects of NEEDLE's training procedure. Sheng et al. (2024) employ inverse-frequency class weighting to address the extreme imbalance between common supernovae and rare transients, with approximately 80× more SN training examples than TDE examples. van den Goorbergh et al. (2022) demonstrated that such corrections systematically inflate minority-class probability estimates, which is precisely the pattern we observe for both SLSN-I (gap = −0.423) and TDE (gap = −0.296).

Global temperature scaling with $T = 1.552$ *worsens* ECE from 0.073 to 0.169 (model-instance). This failure is expected: a single T cannot simultaneously soften overconfident SLSN-I/TDE predictions ($T > 1$ needed) and sharpen underconfident SN predictions ($T < 1$ needed). Kull et al. (2019) documented this fundamental limitation. Per-class temperature scaling yields $T_{\mathrm{SN}} = 1.85$, $T_{\mathrm{SLSN\text{-}I}} = 2.03$, and $T_{\mathrm{TDE}} = 0.54$, confirming the opposing calibration directions, but worsens every individual classwise ECE. These results indicate that, within the range of post-hoc rescaling methods we evaluated and at the present sample sizes (n_{pred} of 39 for SLSN-I and 46 for TDE at the object level), no single global or per-class temperature can simultaneously sharpen the underconfident SN distribution and soften the overconfident rare-class distributions to produce uniform improvement. We therefore characterize the NEEDLE result as strongly suggestive of class-imbalance-induced, class-asymmetric miscalibration that is not corrected by the single-parameter post-hoc rescaling methods evaluated here. Whether more flexible methods, such as Dirichlet calibration (Kull et al. 2019) or focal calibration (Mukhoti et al. 2020), can recover acceptable per-class calibration on a larger labeled rare-class sample remains an open question.

5.4. *Comparative Summary*

The three systems should not be read as a calibration ranking because their evaluation regimes differ. Instead, the comparison identifies distinct failure modes. ALeRCE exhibits correctable underconfidence on the four-class transient subspace, where temperature scaling substantially improves both ECE and high-threshold candidate yield. Fink SuperNNova combines accepted-set miscalibration with class-dependent abstention; formal calibration improvement is achievable but does not translate into high-precision triage utility at operational thresholds. NEEDLE has low aggregate ECE but substantial classwise errors for rare classes that post-hoc methods cannot resolve at current sample sizes.

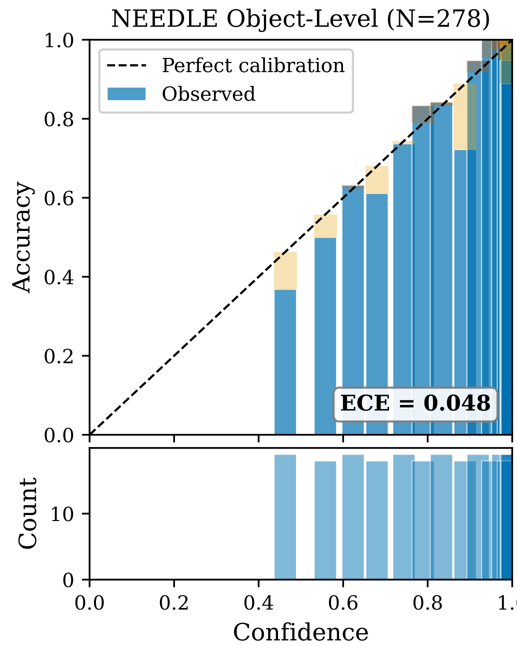

Figure 11. Aggregate reliability diagram for NEEDLE at the object level ($N = 278$ unique objects). ECE = 0.048 places NEEDLE at the boundary of the heuristic "well-calibrated" criterion (ECE < 0.05), but this aggregate view conceals severe per-class miscalibration (Table 4 and Figure 12).

These results demonstrate that a broker calibration audit must report not only aggregate ECE, but also coverage, classwise reliability, and threshold-level selection behavior—metrics that together characterize the operational reliability of probability outputs for spectroscopic follow-up.

6. ROBUSTNESS CHECKS

We performed five robustness checks to verify that primary findings are not artefacts of data processing or methodological choices.

ALeRCE class restriction. Restricting ALeRCE's 15-class output to the 4-class transient subset does not drive the observed miscalibration: the 4-class renormalized ECE (0.271) matches the native 15-class ECE (0.283) within 4%, with negligible per-class probability inflation (1.02–1.06×). The 15-class analysis serves as a harder baseline (more classes, lower accuracy) confirming the 4-class restriction is conservative.

NEEDLE object-level deduplication. The NEEDLE evaluation set contains 429 model-instance predictions across 278 unique objects, with 43 objects appearing in two or more held-out test sets. Deduplicating to object level via mean-of-model-probabilities yields ECE = 0.048 (model-instance: 0.073), a 34% reduction. The 100% inter-model class agreement across all 43 multi-model objects (zero class disagreements) validates the averaging methodology: model variance in predicted class is zero, and confidence uncertainty dominates. Per-class miscalibration patterns persist in both analyses. Object-level results are primary; model-instance results are reported in §5.3.4.

ALeRCE operational gain: held-out cross-validation. The paper's 21× operational gain (20 $\rightarrow$ 425 objects at $p > 0.80$) was originally computed on the full dataset with temperature $T = 0.357$. To address potential data leakage, we recomputed with stratified 5-fold cross-validation: T fitted on four folds, gain measured on the held-out fold. The pooled held-out result (20 $\rightarrow$ 418 objects, gain = 20.9×, precision = 0.907) is essentially identical to the full-dataset claim, because T is a single scalar that converges to 0.357 ± 0.012 regardless of which fold trains it. Per-fold gain shows high variance ($30 \times \pm 23\times$) due to tiny pre-scaling counts per fold (1–6 objects), but the pooled estimate is the appropriate metric (Figure 14).

Fink RF calibration method comparison: not applicable. We initially evaluated temperature scaling, Platt scaling, and isotonic regression on Fink RF's 76 non-zero predictions (5-fold CV) and observed nominal ECE values of 0.175 (T-scaling), 0.198 (Platt), and 0.123 (isotonic). However, because the 76 non-zero objects are an idiosyncratic subset of BTS that happened to pass the early-classifier eligibility gate (Section 3), this comparison does not reflect calibration performance on the intended Fink RF target population. We therefore omit Fink RF calibration-method recommendations from this paper and recommend that future calibration audits of the early-classifier evaluate on alert-stream samples within its design regime. Figure 15 is retained in Appendix B for transparency.

Bin-count sensitivity. ECE was computed at 5, 10, 15, and 20 equal-mass bins for all classifiers (Figure 16). ALeRCE ($N = 1114$) shows zero relative variation (0.0%), Fink SNN ($N = 793$) 6.3%, and Fink RF ($N = 76$) 9.2%, all below the "acceptable" threshold of 10%. NEEDLE ($N = 278$, 3 classes) shows 56.5% relative variation due to small sample size and rare-class imbalance: with 15 bins, only ≈18 objects occupy each bin, and the 15 SLSN-I and 21 TDE true objects span fractional bins. The wide CI $[0.048, 0.111]$ already captures this bin-count uncertainty; the 15-bin point estimate of 0.048 is neither the minimum nor maximum across bin counts (0.036 at 5–10 bins; 0.061 at 20 bins). For large-sample classifiers, 15 bins is well within the stable region.

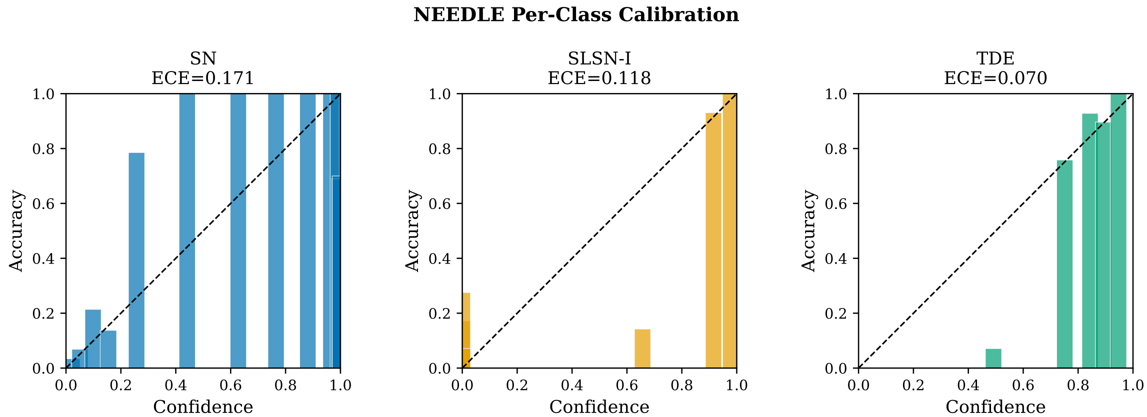

Figure 12. Per-class reliability diagrams for NEEDLE revealing class-asymmetric miscalibration hidden by the aggregate ECE. SN is underconfident (bars above diagonal); SLSN-I is severely overconfident (high-confidence bars below diagonal, especially near 1.0); TDE is overconfident. This asymmetry arises directly from inverse-frequency class weighting during training.

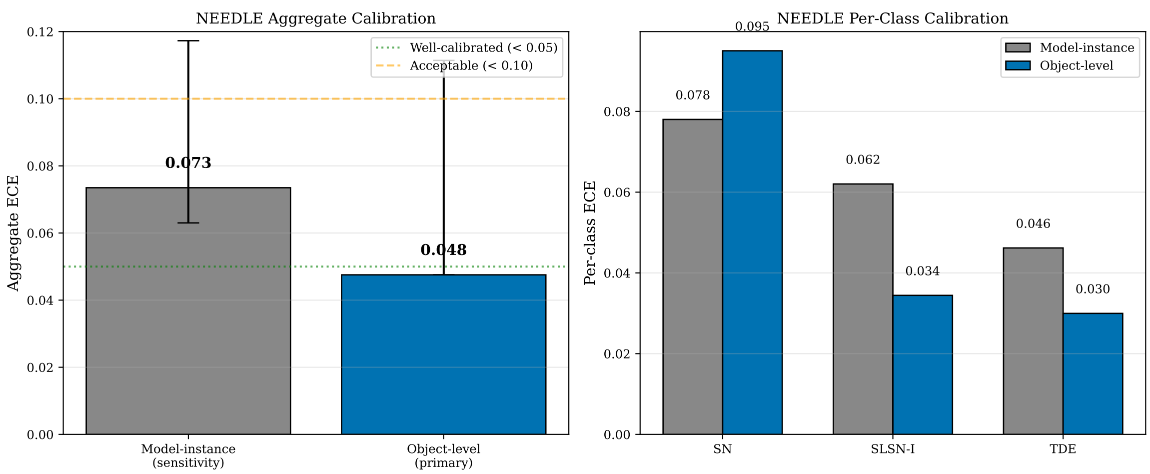

Figure 13. Comparison of NEEDLE calibration at model-instance level ($N = 429$) and object level ($N = 278$, primary). Left: aggregate ECE with 95% bootstrap confidence intervals. Right: per-class ECE for SN, SLSN-I, and TDE. The object-level ECE is lower in aggregate because rare-class objects (which are more miscalibrated) appear in fewer models' test sets when counted once per object.

7. DISCUSSION

7.1. *Calibration Failure Modes Are Architecture- and Training-Dependent*

Production transient classifiers do not share a single calibration failure mode. ALeRCE shows systematic underconfidence consistent with probability compression in ensemble methods (Niculescu-Mizil & Caruana 2005; Boström 2008); Fink SuperNNova behaves as a selective classifier whose calibration must be interpreted conditional on accepted predictions; and NEEDLE demonstrates that aggregate ECE can conceal class-specific failures arising from training imbalance. These differences argue against applying a single generic calibration correction across broker systems and against interpreting aggregate ECE as a sufficient summary of operational reliability.

For ALeRCE, the Balanced Random Forest architecture produces a predictable underconfidence profile: ensemble averaging biases vote fractions toward 0.5, and majority-class undersampling during training further distorts the effective class prior (van den Goorbergh et al. 2022). This profile is correctable by probability sharpening because the miscalibration is approximately uniform across confidence levels. The SLSN exception—mild overconfidence despite the general underconfidence pattern—is directionally consistent with minority-class probability inflation from the Balanced Random Forest architecture, though the small SLSN sample limits the precision of this estimate.

For NEEDLE, the class-asymmetric miscalibration—SN underconfident while SLSN-I and TDE are overconfident—is consistent with the expected effects of inverse-frequency class weighting, which can inflate minority-class probability estimates (van den Goorbergh et al. 2022). Because we do not retrain NEEDLE under alternative weighting schemes, we interpret this as a plausible mechanism rather than a causal ablation result. The structural implication—that single-parameter post-hoc rescaling cannot simultaneously correct opposing miscalibration directions—follows directly from the definition of global temperature scaling.

7.2. *Coverage and Abstention Are Part of Broker Reliability*

The Fink results show that calibration cannot be evaluated independently of coverage. For classifiers with eligibility gates, the absence of a non-zero score is not a low-confidence probability estimate but a classifier abstention. A calibration audit that ignores this distinction can produce misleading ECE values and conceal the operational effect of class-dependent abstention on the accepted-set composition.

This distinction generalizes beyond Fink. Any broker classifier with data-quality thresholds, minimum-epoch requirements, or host-galaxy cuts will exhibit some form of selective prediction. For such classifiers, retrieval, accepted-set coverage, class-conditional abstention rates, and conditional calibration are separate quantities that must be reported independently. The abstention bias framework we apply here—reweighting accepted-set calibration to the target population prior—is a practical tool for quantifying how selective abstention distorts the conditional calibration estimate.

7.3. *Temperature Scaling as a Diagnostic, Not a Universal Fix*

Temperature scaling is useful not only as a post-hoc correction but as a probe of miscalibration structure. The three outcomes we observe illustrate this diagnostic value.

For ALeRCE, a finite $T < 1$ that improves ECE substantially indicates approximately uniform underconfidence correctable by sharpening. For Fink SuperNNova, $T > 1$ improves formal ECE but compresses the score distribution below operational thresholds—a pattern indicating that the classifier's inherent uncertainty is real and that calibration reveals rather than resolves it. For NEEDLE, global temperature scaling worsens aggregate ECE because the miscalibration is class-asymmetric: a single scalar cannot simultaneously sharpen underconfident SN predictions and soften overconfident SLSN-I predictions.

We therefore recommend that broker teams apply temperature scaling not only as a calibration correction but as a first-pass diagnostic of miscalibration structure before committing to more complex post-hoc methods.

7.4. *Rare Classes, Taxonomy, and the Limits of Calibration*

Rare transient classes expose a fundamental boundary of calibration analysis. Calibration can only be evaluated for classes represented in the classifier output space and supported by enough confirmed examples to estimate empirical frequencies reliably. For rare events such as TDEs and SLSNe, the limiting constraint is often sample size rather than classifier quality.

In this study, SLSN-I calibration direction is robust but magnitude is sample-size sensitive; TDE calibration direction is present at the object level but not robustly resolved under the model-instance sensitivity analysis. For ALeRCE, the production taxonomy evaluated here does not include a TDE output class, making calibration undefined for that class in this audit. These cases illustrate that for rare transient science, the relevant operational question is often coverage and taxonomy completeness rather than ordinary ECE. Broker systems targeting rare classes need mechanisms beyond standard classification—such as explicit rare-class flags, anomaly scores, or human-review queues—that are not captured by calibration metrics alone.

7.5. *Operational Implications for Spectroscopic Follow-up*

The operational consequences of miscalibration are asymmetric across systems. For underconfident classifiers such as ALeRCE, miscalibration suppresses candidate yield at high-confidence thresholds: calibration converts a regime where almost no objects pass $p > 0.8$ into one where hundreds of high-precision candidates are available. For overconfident classifiers such as Fink SuperNNova, calibration has the opposite effect: it honestly compresses scores, eliminating high-threshold selections even as it improves formal ECE (Table 3). These contrasting outcomes show that calibration metrics and threshold-level utility curves answer different questions and must both be reported for broker outputs to be operationally trustworthy.

We recommend that broker evaluations report, at minimum: classwise reliability curves alongside aggregate ECE; coverage and class-conditional abstention for classifiers with eligibility gates; and threshold-level selection counts at operationally relevant probability cuts.

7.6. *Implications for the LSST Era*

The specific ECE values reported here are conditional on ZTF alert cadence, the BTS-bright spectroscopic regime, and the classifier versions evaluated. They should not be interpreted as forecasts of Rubin/LSST broker performance. LSST cadence, photometric depth, and evolving broker pipelines will produce different absolute calibration values, and direct quantitative transfer is not warranted.

What transfers is the audit framework itself. The distinction between top-label and classwise calibration, the selective classification framing for abstention-prone classifiers, and the operational threshold analysis are methodology choices that apply regardless of survey. A future audit targeting Rubin alert packets directly would require spectroscopic ground truth from LSST-era surveys, audited classifier versions deployed on LSST alerts, and cadence-aware sensitivity analyses. Until such validation exists, the present work should be regarded as a ZTF-era audit whose findings inform rather than forecast that work.

8. LIMITATIONS

BTS selection function. The ZTF Bright Transient Survey is magnitude-limited and spectroscopically selected. The evaluation set represents a bright, confirmed-transient sample rather than a population-representative sample of all ZTF alerts. Calibration metrics reported here are conditional on this selection.

Classifier-specific evaluation regimes. ALeRCE ECE is a four-class transient-subspace calibration estimate, not a global calibration estimate over the native 15-class taxonomy. Fink SuperNNova ECE is conditional on the accepted non-zero subset and must be interpreted together with coverage and abstention rates. NEEDLE ECE is evaluated on the held-out test objects of publicly released models. These estimands are not directly interchangeable.

Small rare-class samples. For NEEDLE, the SLSN-I sample ($n_{\mathrm{true}} = 15$) and TDE sample ($n_{\mathrm{true}} = 21$) at the object level limit the statistical precision of classwise calibration estimates. Calibration directions are informative; exact magnitudes should be interpreted as strongly suggestive rather than definitive.

ALeRCE retrieval bias. ALeRCE predictions were retrieved for 80.0% of BTS objects, with retrieval significantly class-dependent and biased toward longer-duration transients. ALeRCE calibration metrics are conditional on the retrieval-eligible subset of BTS-bright transients.

Pseudo-logit transformation. Temperature scaling for ALeRCE uses the heuristic transformation $z_k = \log \hat{p}_k$ rather than native logits. The empirical success of this approach does not constitute theoretical justification; alternative post-hoc methods such as Platt scaling or isotonic regression may be equally or more appropriate.

Scope of post-hoc calibration. We evaluate temperature scaling as the primary post-hoc method. More flexible methods such as Dirichlet calibration (Kull et al. 2019) or focal calibration (Mukhoti et al. 2020) may produce better classwise calibration for NEEDLE but require larger labeled rare-class samples to fit reliably (Vaicenavicius et al. 2019).

Snapshot evaluation. All broker predictions were retrieved on 2026 April 1. Classifier versions may have evolved since this date; results characterize the versions evaluated, not permanent assessments of the broker systems.

9. CONCLUSIONS

We present, to our knowledge, the first systematic calibration audit of production transient classifiers deployed on the ZTF alert stream, using spectroscopically confirmed BTS transients as ground truth. The audit shows that broker probability outputs exhibit distinct reliability failure modes not captured by accuracy or aggregate ECE alone. Our main conclusions are:

1. ALeRCE exhibits systematic underconfidence on the evaluated four-class transient subspace. Temperature scaling substantially reduces ECE and increases the number of high-confidence candidates available for threshold-based triage, demonstrating that calibration has direct operational consequences for spectroscopic follow-up allocation.

2. Fink SuperNNova is best interpreted as a selective classifier whose calibration must be evaluated conditionally on the accepted non-zero subset and jointly with class-dependent abstention rates. Formal improvement in ECE after temperature scaling does not necessarily translate into improved high-threshold follow-up utility.

3. NEEDLE demonstrates the limitation of aggregate calibration metrics: its object-level aggregate ECE is low, but classwise analysis reveals rare-class reliability failures hidden by top-label ECE. Temperature scaling worsens calibration in this setting, suggesting that single-parameter post-hoc rescaling is insufficient for class-asymmetric miscalibration.

4. Calibration audits for production broker classifiers should report not only aggregate ECE, but

also classwise reliability curves, coverage and class-conditional abstention, taxonomy limitations, and threshold-level selection behavior. These diagnostics together characterize the operational reliability of probability outputs in a way that no single scalar metric can.

Calibration metrics are not yet routinely reported alongside broker probability outputs, despite their direct relevance to spectroscopic resource allocation. As alert volumes increase in the Rubin/LSST era, broker probabilities will function as decision variables rather than auxiliary classifier outputs, and their reliability must be evaluated accordingly. The calibration diagnostics developed here—classwise reliability, coverage, abstention bias, and threshold-level utility—provide natural inputs to deferral-aware classification frameworks that learn when to route transients to human review rather than automated decision (Madras et al. 2018; Mozannar & Sontag 2020), a direction we pursue in subsequent work.

DATA AND CODE AVAILABILITY

All analysis code, data acquisition scripts, and results are publicly available at https://github.com/pallavi-2000/transient-calibration-audit. The exact version corresponding to this manuscript is tagged as `v1.0-submitted` (commit hash `be9c6f9`). A pinned Python environment is provided in `requirements.txt` (Python 3.8.0; full dependency freeze in `requirements_frozen_full.txt`), and data acquisition timestamps and API endpoints are documented in `DATA_ACQUISITION.md` (all broker predictions retrieved 2026-04-01). Spectroscopic ground truth is drawn from the publicly accessible ZTF Bright Transient Survey (Fremling et al. 2020; Perley et al. 2020). ALeRCE and Fink predictions were obtained via their respective public APIs. NEEDLE predictions were generated using publicly available pre-trained models from Sheng et al. (2024).

We thank Matt Nicholl for encouragement and for suggesting connections with the NEEDLE team, and Xinyue Sheng for making the NEEDLE models and dataset publicly available. This work made use of the ALeRCE broker (Förster et al. 2021), the Fink broker (Möller et al. 2021), and data from the ZTF Bright Transient Survey (Fremling et al. 2020; Perley et al. 2020). This research has made use of NASA's Astrophysics Data System.

Software: `NumPy` (Harris et al. 2020), `SciPy` (Virtanen et al. 2020), `pandas` (McKinney 2010), `Matplotlib` (Hunter 2007), `TensorFlow` (Abadi et al. 2016)

REFERENCES

Abadi, M., et al. 2016, in Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation, 265–283

Bellm, E. C., et al. 2019, PASP, 131, 018002, doi: 10.1088/1538-3873/aaecbe

Boström, H. 2008, in 2008 Seventh International Conference on Machine Learning and Applications (IEEE), 121–126, doi: 10.1109/ICMLA.2008.107

Brier, G. W. 1950, Monthly Weather Review, 78, 1

Dawid, A. P. 1982, Journal of the American Statistical Association, 77, 605

de Soto, K. D., et al. 2024, ApJ, 974, 169, doi: 10.3847/1538-4357/ad6869

Fang, L., & Orabona, F. 2020, in IEEE International Conference on Pattern Recognition

Förster, F., et al. 2021, AJ, 161, 242, doi: 10.3847/1538-3881/abe9bc

Fremling, C., et al. 2020, ApJ, 895, 32, doi: 10.3847/1538-4357/ab8943

Geifman, Y., & El-Yaniv, R. 2017, in Advances in Neural Information Processing Systems, Vol. 30 (Curran Associates, Inc.), 4878–4887

Gneiting, T., & Raftery, A. E. 2007, Journal of the American Statistical Association, 102, 359, doi: 10.1198/016214506000001437

Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. 2017, in Proceedings of the 34th International Conference on Machine Learning (PMLR), 1321–1330

Harris, C. R., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi: 10.1109/MCSE.2007.55

Ivezić, Ž., et al. 2019, ApJ, 873, 111, doi: 10.3847/1538-4357/ab042c

Kull, M., Perello-Nieto, M., Kängsepp, M., et al. 2019, in Advances in Neural Information Processing Systems, Vol. 32

Leoni, M., Ishida, E. E. O., Peloton, J., & Möller, A. 2022, A&A, 663, A13, doi: 10.1051/0004-6361/202142715

Madras, D., Pitassi, T., & Zemel, R. 2018, in Advances in Neural Information Processing Systems, Vol. 31 (Curran Associates, Inc.), 6147–6157

McKinney, W. 2010, in Proceedings of the 9th Python in Science Conference, 56–61

Möller, A., & de Boissière, T. 2020, MNRAS, 491, 4277, doi: 10.1093/mnras/stz3312

Möller, A., et al. 2021, MNRAS, 501, 3272, doi: 10.1093/mnras/staa3602

Mozannar, H., & Sontag, D. 2020, in Proceedings of Machine Learning Research, Vol. 119, Proceedings of the 37th International Conference on Machine Learning (PMLR), 7076–7087

Mukhoti, J., Kulharia, V., Sanber, A., Torr, P. H. S., & Dokania, P. K. 2020, in Advances in Neural Information Processing Systems, Vol. 33

Naeini, M. P., Cooper, G. F., & Hauskrecht, M. 2015, in Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 29, 2901–2907

Narayan, G., et al. 2018, ApJS, 236, 9, doi: 10.3847/1538-4365/aab781

Niculescu-Mizil, A., & Caruana, R. 2005, in Proceedings of the 22nd International Conference on Machine Learning (ACM), 625–632

Nixon, J., Dusenberry, M. W., Zhang, L., Jerfel, G., & Tran, D. 2019, in CVPR Workshops, 38–41

Perley, D. A., et al. 2020, ApJ, 904, 35, doi: 10.3847/1538-4357/abbd98

Roelofs, R., Cain, N., Shlens, J., & Mozer, M. C. 2022, in Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (PMLR)

Sánchez-Sáez, P., et al. 2021, AJ, 161, 141, doi: 10.3847/1538-3881/abd5c1

Sheng, X., et al. 2024, MNRAS, 531, 2474, doi: 10.1093/mnras/stae1253

Smith, K. W., et al. 2019, Research Notes of the AAS, 3, 26, doi: 10.3847/2515-5172/ab020f

Tung, A. 2025, arXiv e-prints. https://arxiv.org/abs/2511.00614

Vaicenavicius, J., Widmann, D., Andersson, C., et al. 2019, arXiv e-prints. https://arxiv.org/abs/1902.06977

van den Goorbergh, R., van Smeden, M., Timmerman, D., & Zwinderman, A. H. 2022, Journal of the American Medical Informatics Association, 29, 1525, doi: 10.1093/jamia/ocac093

Virtanen, P., et al. 2020, Nature Methods, 17, 261, doi: 10.1038/s41592-019-0686-2

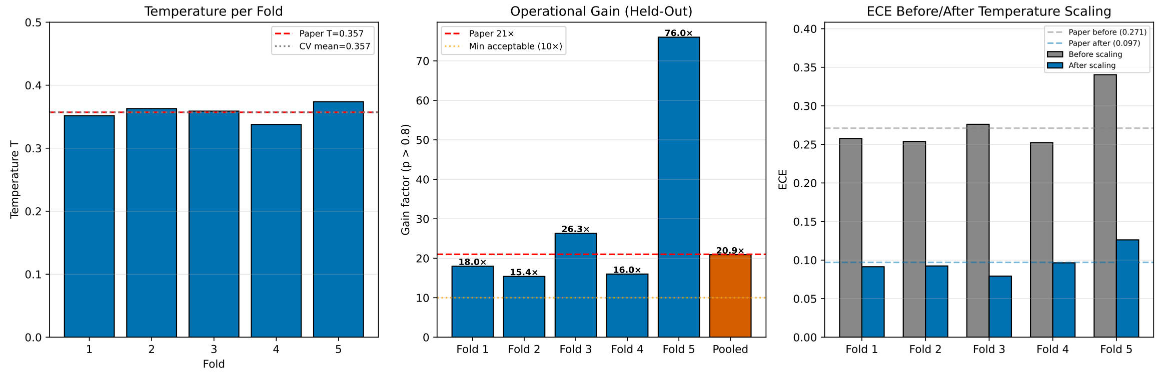

Figure 14. Cross-validated operational gain for ALeRCE temperature scaling (5-fold stratified CV). *Left*: temperature T per fold; mean $T = 0.357 \pm 0.012$ is virtually identical to the full-dataset estimate, confirming that the scalar calibration parameter is insensitive to training-set composition and that no data leakage inflates the paper's operational gain claim. *Centre*: per-fold and pooled gain factor at $p > 0.80$; the pooled held-out result ($20 \rightarrow 418$ objects, gain $= 20.9\times$) matches the paper claim of $21\times$. High per-fold variance (15–76$\times$) arises because the pre-scaling count per fold is 1–6 objects. *Right*: per-fold ECE before (0.276 ± 0.033) and after (0.097 ± 0.016) temperature scaling, consistent with the full-dataset improvement.

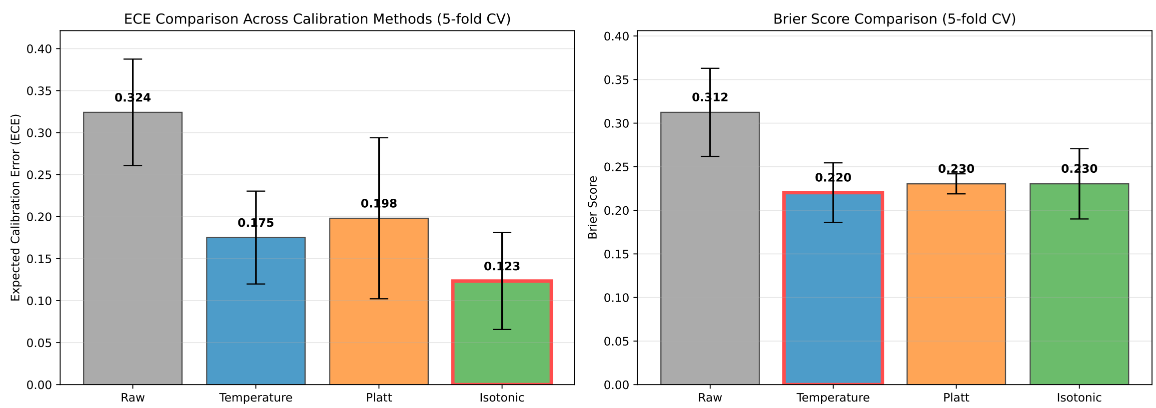

Figure 15. Comparison of three post-hoc calibration methods for Fink RF non-zero predictions ($N = 76$, 5-fold CV). Temperature scaling ($\mathrm{ECE} = 0.175 \pm 0.055$), Platt scaling ($\mathrm{ECE} = 0.198 \pm 0.096$), and isotonic regression ($\mathrm{ECE} = 0.123 \pm 0.058$) all reduce ECE from the raw value of 0.324. Isotonic regression achieves the lowest point estimate but with the highest variance; the overlapping uncertainty ranges and risk of overfitting on $\approx$12 test objects per fold make temperature scaling the more conservative recommendation at $N = 76$.

APPENDIX

A. CROSS-VALIDATED OPERATIONAL GAIN (ALERCE)

B. CALIBRATION METHOD COMPARISON FOR FINK RF

C. BIN-COUNT SENSITIVITY ANALYSIS

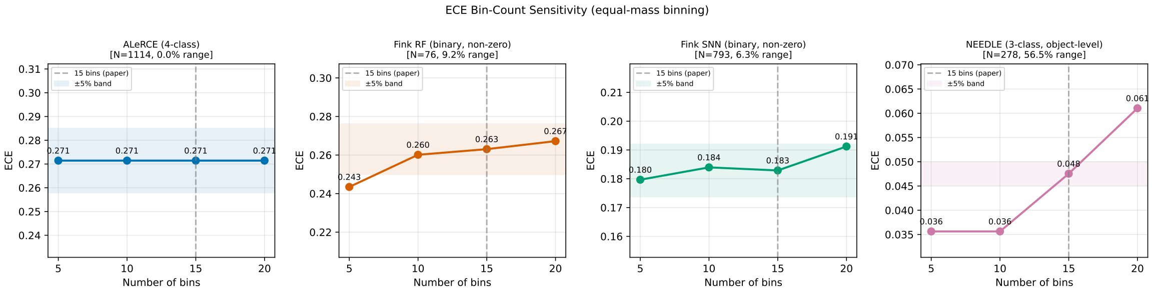

Figure 16. ECE as a function of bin count (5, 10, 15, 20 equal-mass bins) for all classifiers. ALeRCE ($N = 1114$): 0.0% relative variation (excellent). Fink RF ($N = 76$): 9.2% (acceptable). Fink SNN ($N = 793$): 6.3% (acceptable). NEEDLE ($N = 278$): 56.5% (poor, driven by small sample and rare-class imbalance). The dashed vertical line marks the paper's choice of 15 bins; it is neither a minimum nor maximum for any classifier, and the choice is well within the stable region for ALeRCE and Fink SNN. For NEEDLE, the wide bootstrap CI $[0.048, 0.111]$ already subsumes the bin-count uncertainty.